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.NET INTERVIEW QUESTIONS

The complete guide for freshers — covering C#, OOP, ASP.NET Core, Entity Framework, LINQ, Web API and more.

WITH ANSWERS • FOR FRESHERS

Q U E S T I O N S A N D A N S W E R S

1 What is .NET Framework?

A. A software development platform by Microsoft for building and running Windows applications, providing a Common Language Runtime (CLR) and a large class library (FCL).

2 What is .NET Core / .NET (5+)?

A. A free, open-source, cross-platform successor to .NET Framework that runs on Windows, Linux, and macOS, and is faster and modular.

3 What is CLR (Common Language Runtime)?

A. The execution engine of .NET that manages memory, security, exception handling, garbage collection, and JIT compilation of code.

4 What is CTS (Common Type System)?

A. A standard that defines how types are declared, used, and managed in the CLR, enabling cross-language integration in .NET.

5 What is CLS (Common Language Specification)?

A. A set of rules that all .NET languages must follow so that code written in one language (C#, VB.NET) can be used from another.

6 What is MSIL / IL code?

A. Microsoft Intermediate Language is the CPU-independent code generated by .NET compilers, which the JIT compiler converts to native machine code at runtime.

7 What is JIT compilation?

A. Just-In-Time compilation converts IL code into native machine code at runtime, just before it is executed, improving performance on the target machine.

8 What is an assembly in .NET?

A. A compiled unit of deployment (.dll or .exe) containing IL code, metadata, and a manifest that describes the assembly's contents and version.

9 What is the Global Assembly Cache (GAC)?

A. A machine-wide cache used to store assemblies that are shared by multiple applications on a Windows machine.

10 What is the difference between managed and unmanaged code?

A. Managed code runs under the CLR, which handles memory and security; unmanaged code runs directly on the OS and the developer manages memory manually.

11 What is garbage collection in .NET?

A. An automatic memory management feature where the CLR reclaims memory occupied by objects that are no longer reachable or referenced by the application.

12 What are the different types of application in .NET?

A. Console apps, Windows Forms, WPF, ASP.NET Web apps, Web APIs, Windows Services, and cross-platform apps built with .NET MAUI or Blazor.

13 What is NuGet?

A. A package manager for .NET that lets developers create, share, and consume useful code packages (libraries) in their projects.

14 What is the difference between .NET Framework and .NET Core?

A. .NET Framework is Windows-only and no longer actively enhanced, while .NET Core (now .NET) is open-source, cross-platform, and has better performance.

15 What is Object-Oriented Programming (OOP)?

A. A programming paradigm based on objects that combine data (fields) and behavior (methods), built around encapsulation, inheritance, polymorphism, and abstraction.

16 What is a class and an object?

A. A class is a blueprint/template that defines properties and methods, while an object is an instance created from that class at runtime.

17 What is encapsulation?

A. The practice of bundling data and methods together inside a class and restricting direct access to internal state using access modifiers like private.

18 What is inheritance?

A. A mechanism where a class (derived/child) acquires the properties and behaviors of another class (base/parent), promoting code reuse.

19 What is polymorphism?

A. The ability of an object to take many forms, allowing the same method call to behave differently based on the object type, via overriding or overloading.

20 What is abstraction?

A. The concept of hiding complex implementation details and showing only the essential features of an object, typically via abstract classes or interfaces.

21 What is the difference between an abstract class and an interface?

A. An abstract class can have implementation and state and supports single inheritance; an interface only declares members (with C# 8+ allowing default methods) and supports multiple inheritance.

22 What is method overloading?

A. Defining multiple methods with the same name but different parameter lists (number, type, or order) within the same class.

23 What is method overriding?

A. Redefining a base class's virtual or abstract method in a derived class using the override keyword to provide specific behavior.

24 What is the difference between overloading and overriding?

A. Overloading is compile-time polymorphism with the same method name and different signatures; overriding is runtime polymorphism that redefines an inherited method.

25 What is a constructor?

A. A special method automatically invoked when an object is created, used to initialize the object's fields; it has the same name as the class and no return type.

26 What is a static class and static member?

A. A static class cannot be instantiated and contains only static members; static members belong to the class itself rather than any specific instance.

27 What is the 'sealed' keyword used for?

A. It prevents a class from being inherited further, or prevents an overridden method from being overridden again in derived classes.

28 What is a virtual method?

A. A method in a base class marked with the virtual keyword, allowing it to be overridden by derived classes using the override keyword.

29 What are value types and reference types?

A. Value types (int, struct, bool) store data directly on the stack, while reference types (class, string, array) store a reference to data on the heap.

30 What is boxing and unboxing?

A. Boxing converts a value type to an object (reference type) by wrapping it on the heap; unboxing extracts the value type back from the object.

31 What is the difference between 'const' and 'readonly'?

A. const is a compile-time constant assigned at declaration, while readonly can be assigned at declaration or in a constructor and is resolved at runtime.

32 What is the 'var' keyword?

A. It lets the compiler infer the variable's type from the assigned value at compile time; the type is still strongly typed, just implicitly declared.

33 What is nullable type in C#?

A. A value type suffixed with '?' (e.g., int?) that can hold a normal value or null, useful for representing optional or missing data.

34 What is the difference between '==' and '.Equals()'?

A. '==' compares references for reference types by default (unless overloaded), while '.Equals()' can be overridden to compare actual content/value equality.

35 What are delegates in C#?

A. Type-safe function pointers that hold references to methods with a matching signature, allowing methods to be passed as parameters or invoked dynamically.

36 What are events in C#?

A. A mechanism built on delegates that allows a class to notify subscribers when something happens, following the publisher-subscriber pattern.

37 What are lambda expressions?

A. Concise anonymous functions using the ' $\Rightarrow$ ' syntax, commonly used with delegates, LINQ queries, and functional-style code.

38 What is the difference between 'out' and 'ref' parameters?

A. 'ref' requires the variable to be initialized before being passed; 'out' does not require prior initialization but must be assigned inside the method.

39 What are extension methods?

A. Static methods that let you add new functionality to an existing type without modifying its source code, defined using the 'this' keyword on the first parameter.

40 What is the difference between 'string' and 'StringBuilder'?

A. 'string' is immutable, so every modification creates a new object; 'StringBuilder' is mutable and more efficient for repeated string modifications.

41 What is async/await used for?

A. Keywords that enable asynchronous, non-blocking programming, allowing a method to perform long-running work without blocking the calling thread.

42 What is the difference between 'Task' and 'Thread'?

A. A Thread is a low-level OS construct for parallel execution, while a Task is a higher-level abstraction from the Task Parallel Library for async work, often reusing thread pool threads.

43 What are generics in C#?

A. A feature that allows classes, methods, and interfaces to operate on a type specified at usage time, providing type safety and reusability (e.g., `List<T>`).

44 What is pattern matching in C#?

A. A feature (via 'is', 'switch' expressions) that lets you test a value against a pattern like type, shape, or property, and extract data concisely.

45 What is the 'yield' keyword used for?

A. It is used inside an iterator method to return elements one at a time lazily, simplifying implementation of custom collections and IEnumerable.

46 What is the difference between an Array and a List in C#?

A. An array has a fixed size set at creation, while a List<T> is dynamic and can grow or shrink automatically as elements are added or removed.

47 What is the difference between IEnumerable and IQueryable?

A. IEnumerable executes queries in memory and is best for LINQ to Objects; IQueryable builds an expression tree and can execute queries on the data source, like a database.

48 What is LINQ?

A. Language Integrated Query is a set of features that lets you write SQL-like queries directly in C# to filter, sort, and transform collections and data sources.

49 What is the difference between List and Dictionary?

A. A `List<T>` stores an ordered collection of values accessed by index, while a `Dictionary<TKey,TValue>` stores key-value pairs accessed efficiently by unique key.

50 What is the difference between Array and ArrayList?

A. An Array is strongly typed and fixed-size, while ArrayList is a legacy non-generic collection that stores objects and requires boxing/unboxing for value types.

51 What is a HashSet in C#?

A. A collection that stores unique elements with no duplicates, offering fast lookups, additions, and removals using a hash-based structure.

52 What is deferred execution in LINQ?

A. A behavior where a LINQ query is not executed when defined, but only when the results are actually iterated over, such as in a foreach loop.

53 What is the difference between First() and FirstOrDefault() in LINQ?

A. First() throws an exception if no element is found, while FirstOrDefault() returns the type's default value (like null or 0) instead of throwing.

54 What is the difference between Select and SelectMany in LINQ?

A. Select projects each element into a new form producing one output per input, while SelectMany flattens nested collections into a single sequence.

55 What is the difference between Queue and Stack?

A. A Queue follows First-In-First-Out (FIFO) ordering, while a Stack follows Last-In-First-Out (LIFO) ordering for adding and removing elements.

56 What is exception handling in C#?

A. A mechanism using try, catch, finally, and throw blocks to detect and handle runtime errors gracefully without crashing the application.

57 What is the purpose of the 'finally' block?

A. Code inside 'finally' always executes after try/catch, whether or not an exception occurred, commonly used to release resources like files or connections.

58 What is a custom exception?

A. A user-defined exception class that inherits from the Exception class, used to represent application-specific error conditions.

59 What is the difference between 'throw' and 'throw ex'?

A. 'throw' preserves the original stack trace of the exception, while 'throw ex' resets the stack trace, making debugging harder.

60 What is the 'using' statement used for?

A. It ensures that an IDisposable object is properly disposed of (resources released) once it goes out of scope, even if an exception occurs.

61 What is the IDisposable interface?

A. An interface with a Dispose() method that classes implement to release unmanaged resources like file handles, database connections, or network sockets.

62 What is the difference between Finalize and Dispose?

A. Dispose is called explicitly by the developer to release resources immediately, while Finalize is called by the garbage collector before reclaiming an object's memory.

63 What are the generations in .NET Garbage Collection?

A. Generation 0 (short-lived objects), Generation 1 (medium-lived), and Generation 2 (long-lived objects); the GC optimizes collection by focusing more on Gen 0.

64 What is multithreading?

A. A technique that allows multiple threads to run concurrently within a single process, improving performance for parallel and responsive applications.

65 What is a deadlock?

A. A situation where two or more threads are blocked forever, each waiting for a resource held by the other, so none can proceed.

66 What is the lock keyword used for?

A. It ensures that only one thread can access a critical section of code at a time, preventing race conditions on shared resources.

67 What is reflection in .NET?

A. A feature that allows inspecting and interacting with metadata of types, methods, and assemblies at runtime, enabling dynamic object creation and invocation.

68 What is ASP.NET Core?

A. A cross-platform, high-performance, open-source framework for building modern web apps, APIs, and microservices, and the successor to ASP.NET.

69 What is the MVC pattern?

A. Model-View-Controller separates an application into the Model (data/logic), View (UI), and Controller (handles requests and coordinates Model and View).

70 What is Middleware in ASP.NET Core?

A. Software components assembled into a request pipeline that handle requests and responses, such as authentication, routing, and logging.

71 What is Dependency Injection (DI)?

A. A design pattern where an object's dependencies are provided externally rather than created internally, promoting loose coupling and testability.

72 What are the types of DI service lifetimes in ASP.NET Core?

A. Transient (new instance every time), Scoped (one instance per request), and Singleton (one instance for the application's lifetime).

73 What is Razor syntax?

A. A markup syntax that lets you embed C# code within HTML in .cshtml files to generate dynamic web content in ASP.NET applications.

74 What is routing in ASP.NET Core?

A. The process of mapping incoming HTTP request URLs to the appropriate controller actions or endpoints in the application.

75 What is the difference between ViewData, ViewBag, and TempData?

A. ViewData and ViewBag pass data from controller to view for a single request; TempData persists data across one redirect using session-like storage.

76 What is model binding?

A. A feature that automatically maps data from HTTP requests (form fields, query strings, route data) into action method parameters or model objects.

77 What is model validation?

A. The process of checking that submitted data meets defined rules (using data annotations like [Required]) before it is processed by the application.

78 What are filters in ASP.NET Core MVC?

A. Attributes that run custom logic before or after specific stages of the request pipeline, such as Authorization, Action, Result, and Exception filters.

79 What is the wwwroot folder used for?

A. The default web root folder in ASP.NET Core that stores static files like CSS, JavaScript, and images, served directly to clients.

80 What is Kestrel?

A. A lightweight, cross-platform web server built into ASP.NET Core that is used to host and serve web applications.

81 What is the difference between Authentication and Authorization?

A. Authentication verifies who a user is (identity), while authorization determines what an authenticated user is allowed to do (permissions).

82 What is ADO.NET?

A. A data access technology in .NET that provides classes like SqlConnection, SqlCommand, and DataReader for interacting with relational databases.

83 What is the difference between DataReader and DataSet?

A. DataReader provides fast, forward-only, read-only access to data, while DataSet stores data in memory and supports disconnected, offline data operations.

84 What is Entity Framework (EF)?

A. An Object-Relational Mapper (ORM) for .NET that lets developers work with databases using C# objects instead of writing raw SQL queries.

85 What is the difference between EF Core and EF6?

A. EF Core is the modern, lightweight, cross-platform version of Entity Framework, while EF6 is the older, Windows-focused, more mature version.

86 What is Code First approach in EF?

A. A development approach where you define C# model classes first, and Entity Framework generates the database schema based on those classes.

87 What is a Migration in Entity Framework?

A. A feature that tracks changes to your model classes and applies corresponding incremental updates to the database schema over time.

88 What is lazy loading vs eager loading in EF?

A. Lazy loading fetches related data only when it's accessed, while eager loading (using Include()) fetches related data upfront in the initial query.

89 What is connection pooling?

A. A technique where database connections are reused from a pool rather than opened and closed for every request, improving performance.

90 What is the difference between DbContext and DbSet?

A. DbContext represents a session with the database for querying and saving data, while DbSet<T> represents a collection/table of a specific entity type.

91 What is a Web API?

A. A framework for building HTTP-based services that can be consumed by various clients like browsers, mobile apps, and other applications, typically returning JSON.

92 What is REST and RESTful API?

A. REST is an architectural style using stateless HTTP methods (GET, POST, PUT, DELETE) on resources; a RESTful API follows these principles for web services.

93 What is the difference between PUT and PATCH?

A. PUT replaces an entire resource with the given data, while PATCH applies a partial update, modifying only the specified fields.

94 What are HTTP status codes commonly used in Web APIs?

A. 200 OK, 201 Created, 204 No Content, 400 Bad Request, 401 Unauthorized, 404 Not Found, and 500 Internal Server Error are commonly used codes.

95 What is Swagger/OpenAPI?

A. A tool and specification for documenting and testing Web APIs, automatically generating interactive API documentation from your code.

96 What is JWT (JSON Web Token)?

A. A compact, self-contained token format used for securely transmitting user identity and claims between parties, commonly used for API authentication.

97 What is CORS?

A. Cross-Origin Resource Sharing is a security feature that controls whether a web page from one origin can request resources from a different origin.

98 What is the difference between .NET 6, 7, and 8?

A. .NET 6 was the first unified Long-Term Support (LTS) release; .NET 7 added performance and feature updates; .NET 8 is a newer LTS release with further performance and cloud-native improvements.

99 What is Blazor?

A. A framework for building interactive web UIs using C# instead of JavaScript, supporting both server-side (Blazor Server) and client-side (Blazor WebAssembly) hosting.

100 What is the difference between a monolithic and microservices architecture?

A. A monolith is a single deployable unit containing all functionality, while microservices split an application into small, independently deployable services.